

ResearchMind AI Evaluation Paper

Abstract

This paper evaluates a retrieval augmented generation assistant for research workflows. The system extracts text from uploaded PDF papers, splits content into semantic chunks, creates embeddings, stores vectors, and answers user questions with citations.

Methodology

The prototype uses a FastAPI backend, PostgreSQL metadata storage, FAISS vector indexing, and Gemini language models. Users authenticate, upload PDF files, create chat sessions, ask questions, generate summaries, and compare multiple papers.

Findings

Citation grounded answers reduce hallucination risk by tying responses to retrieved chunks. Study tools such as summaries, flashcards, and literature reviews help researchers review papers faster.

Limitations

The quality depends on PDF text extraction, embedding quality, retrieval configuration, and the availability of the external Gemini API.